MODULE 04 · SECTION 1 - FUNDAMENTALS OF GEN AI

Data: types, labels and the six marks of quality

What the data looks like, whether it carries answers, and whether it is fit to use.

SECTION 1 of 4 EXAM WEIGHT ~30% MODULE 4 of 33

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The cleverest model still reflects its data.
Garbage in, garbage out never retired.

Structured, semi-structured, unstructured

TYPE	WHAT IT LOOKS LIKE	EXAMPLES
Structured	Rows and columns, fixed schema	Orders table, sales ledger
Semi-structured	Tags or keys, flexible shape	JSON, XML, log files
Unstructured	No predefined model	Documents, images, audio, video

ON THE EXAM

Free text written by people is unstructured, even when it is stored in a database column.

Why unstructured data matters so much now

- **Most business knowledge** lives in documents, emails, chats, images and recordings.
- **Traditional ML** worked best on tidy tables, so that knowledge went unused.
- **Foundation models** were trained on unstructured data and read it natively.
- **Multimodal models** handle images, audio and video, not just text.
- **It still needs care:** duplicates, old versions and personal data hide in it.

> THE TAKEAWAY
>
> Generative AI's big business shift is making unstructured data usable without first turning it into tables.

Where each kind of data lives on Google Cloud

GOOGLE CLOUD SERVICE	TYPICAL DATA
BigQuery	Structured and semi-structured data for analysis, at any scale
Cloud SQL and other databases	Live operational records, such as orders and accounts
Cloud Storage	Files of any kind: documents, images, audio, video
Pub/Sub	Streams of events arriving in real time, such as stock updates

REMEMBER

Module 6 follows this data through the ML lifecycle. For now, match the data type to its home.

Labelled or unlabelled?

LABELLED DATA

Each example carries its answer

- Photo tagged sofa or armchair
- Loan marked repaid or defaulted
- Needed for supervised learning
- Costly: people create the labels

UNLABELLED DATA

Raw, with no answer attached

- A folder of untagged photos
- A year of call transcripts
- Used by unsupervised learning
- Plentiful and cheap to collect

THE TIEBREAKER

Labels are what supervised learning needs, and they are usually the slow, expensive part.

Label thousands, or prompt with a few?

TRADITIONAL SUPERVISED MODEL

Thousands of labels

- People tag thousands of examples
- A training project before any results
- Best for high-volume, specialised tasks
- Accuracy measured against the labels

FOUNDATION MODEL

A clear prompt

- Describe the categories in plain words
- Add a few examples if needed
- Results the same day, no training
- Still check a sample by hand

THE TIEBREAKER

If a foundation model can do the task from a prompt, the labelling bill may disappear. Check that first.

The six marks of quality in the exam guide

MARK	THE QUESTION A LEADER ASKS
Completeness	Is anything missing: values, records, whole groups?
Consistency	Is the same thing recorded the same way everywhere?
Relevance	Does this data actually fit the problem, and is it current?
Availability	Can the system reach it, when and where it is needed?
Cost	What will it cost to collect, label, store and maintain?
Format	Is it in a shape the tools can use?

ON THE EXAM

Learn these six by name. The exam can describe a symptom and ask which characteristic has failed.

Wrenfield fails all six

MARK	WHAT WENT WRONG AT WRENFIELD
Completeness	6% of products have no dimensions recorded
Consistency	One colour stored as Sage, sage green and SG
Relevance	An old staff handbook still answers today's questions
Availability	Policies locked in a team folder the AI cannot reach
Cost	Hand-tagging every product photo would take months
Format	Supplier invoices arrive as scanned images

THE TIEBREAKER

Every one of these would make a gen AI project look worse than it is. Fix the data before blaming the model.

The model inherits its data's flaws

- **Data dependency** is a named limitation: outputs depend on training data.
- **Skewed data** gives skewed results, such as a model trained mostly on one group.
- **Dated data** gives dated results, such as old language or old prices.
- **Missing groups** are invisible to the model, and it will not tell you.
- **The fix starts with the data:** broader, current, representative sources.

> ON THE EXAM
>
> When a scenario blames biased or outdated training data, the limitation is data dependency.

More data, or better data?

MORE OF THE SAME

Volume alone

- Repeats existing gaps and biases
- Costs more to store and process
- Can bury the signal in noise
- Feels like progress

BETTER DATA

Quality first

- Covers the groups and cases that matter
- Consistent, current and complete
- Smaller can beat bigger
- Usually the cheaper win

THE TIEBREAKER

If the problem is a gap or a bias, adding more of the same data does not fix it.

What data decisions mean for the business

- **Structured data** is ready for dashboards and predictive models today.
- **Unstructured data** needs extraction, such as Document AI, or a multimodal model.
- **Labels** are a budget line and a timeline, not a detail.
- **Personal data** needs protecting before it is used. Module 27 covers how.
- **Ownership:** someone must keep each source current and correct.

> THE TAKEAWAY
>
> Many gen AI projects stall on data access and quality, not on the model. Plan for the data first.

> Wrenfield wants to understand why customers return sofas. Four data sources are available.
>
> The team wants the source that is unstructured and most likely to explain the reasons in the customer's own words.

Which source should they choose?

A Free-text return reasons typed by customers

B A returns table with date, product code and refund amount

C Stock levels by warehouse, exported as CSV

D Delivery timestamps from the courier's system

◆ PAUSE AND DECIDE

Which source should they choose?

A Free-text return reasons typed by customers

Unstructured, and written by customers in their own words.

B A returns table with date, product code and refund amount

Structured. It records what was returned, not why.

C Stock levels by warehouse, exported as CSV

Structured, and unrelated to the reasons.

D Delivery timestamps from the courier's system

Structured timings. Hints at delays, but not in customers' words.

THE SIGNAL

Free text written by people is unstructured, and it holds the why.

> A shopper searches Wrenfield's site for sage green cushions and sees only a third of the matching stock.
>
> The product data records the same colour as Sage, sage green and SG, depending on who entered it.

Which data quality characteristic has failed?

A Completeness

B Relevance

C Availability

D Consistency

◆ PAUSE AND DECIDE

Which data quality characteristic has failed?

A **Completeness**
Nothing is missing. Every product has a colour.

B **Relevance**
Colour is exactly the right data for this search.

C **Availability**
The search can reach the data. It just cannot match it.

D **Consistency**
One value recorded three different ways.

THE SIGNAL

The same thing recorded different ways is a consistency failure.

> A pilot assistant answers staff questions about the returns policy. Its answers are fluent and well organised.
>
> Staff then notice the answers match the 2019 policy. The assistant was pointed at the policy archive, not the current handbook.

Which data quality characteristic failed?

A Format

B Cost

C Completeness

D Relevance

◆ PAUSE AND DECIDE

Which data quality characteristic failed?

A **Format**

The documents were readable. The answers were fluent.

B **Cost**

Nothing here is about the expense of the data.

C **Completeness**

The archive was complete. It was the wrong archive.

D **Relevance**

Out-of-date sources do not fit today's question.

THE SIGNAL

Fluent but wrong because the source is old: a relevance failure, and a warning about confident answers.

> A furniture supplier plans to train its own image classifier to spot items damaged in transit. It has 120,000 delivery photos, and none is tagged as damaged or undamaged.
>
> The project budget is tight, and the director wants to know the main data issue to plan for.

What should the director plan for?

A Time and money to label enough photos for supervised learning

B Nothing extra, because untagged photos can train a supervised model

C Converting the photos into a structured spreadsheet first

D Dropping the photos, because ML cannot use unstructured data

◆ PAUSE AND DECIDE

What should the director plan for?

A Time and money to label enough photos for supervised learning

Damaged or not is a supervised task, and no labels exist yet.

B Nothing extra, because untagged photos can train a supervised model

Supervised learning needs labelled examples.

C Converting the photos into a structured spreadsheet first

Image models learn from images directly.

D Dropping the photos, because ML cannot use unstructured data

Images are a core use of ML and multimodal models.

THE SIGNAL

No labels plus a supervised goal: the labels are the cost line.

Module 4 in five lines

- Structured is rows and columns; unstructured is documents, images, audio, video and free text.
- Labelled data carries the answer and costs effort; unlabelled data is plentiful.
- Six marks: completeness, consistency, relevance, availability, cost, format.
- Gen AI makes unstructured data usable, but the data still sets the ceiling.
- Biased or outdated data gives biased or outdated output: data dependency.